

MANDEEP CHAUHAN

Python Backend Developer | Data Science & Machine Learning

Python · Django · FastAPI · DRF · Flask · REST APIs · SQL · PostgreSQL · MySQL · Redis · Docker · Pandas · NumPy · Scikit-learn · Streamlit
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PROFESSIONAL SUMMARY

Python Backend Developer with 3+ years of professional experience architecting RESTful APIs and production-grade backend systems using Python, FastAPI, Django, Django REST Framework (DRF), and Flask. Proficient in SQL-based database design, query optimisation, and caching strategies across PostgreSQL, MySQL, and Redis, with hands-on expertise in JSON Web Token (JWT) authentication, Role-Based Access Control (RBAC), Swagger/OpenAPI documentation, and containerised deployment via Docker. Complements core backend engineering with applied Data Science and Machine Learning capability developed through hands-on projects spanning Exploratory Data Analysis (EDA), feature engineering, regression and classification modelling, and end-to-end ML application deployment using Scikit-learn, Pandas, NumPy, and Streamlit. Delivers reliable, scalable backend features in Agile/SDLC environments, with hands-on project experience in Data Science and Machine Learning.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, HTML5, CSS3, PHP

Backend Frameworks: FastAPI, Django, Django REST Framework (DRF), Flask, Laravel

Backend & APIs: REST API Design, RESTful APIs, JWT Authentication, RBAC, Swagger/OpenAPI, Postman, JSON, HTTP, Third-Party API Integration

Databases: PostgreSQL, MySQL, Redis, SQLite

Backend Concepts: OOP, MVC Architecture, CRUD, API Documentation, Database Optimization, Query Optimization, Caching, Performance Optimization

DevOps & Tools: Docker, Git, GitHub, Linux, VS Code, Jupyter Notebook

Data Science & ML: Pandas, NumPy, Scikit-learn, XGBoost, Matplotlib, Seaborn, Streamlit, EDA, Feature Engineering, Regression, Classification, Hyperparameter Tuning, Predictive Modeling, Statistical Analysis

PROFESSIONAL EXPERIENCE

Backend Developer / PHP Developer

Apr 2025 – Jul 2026

Spirant Communication Pvt. Ltd. | Noida, Uttar Pradesh

- Developed and maintained ~5 PHP-based web applications supporting core business operations in a live production environment, reducing critical downtime incidents by ~20%.
- Optimised MySQL database queries and schemas across ~10 production tables, improving data retrieval efficiency by ~30% and reducing average query execution time.
- Designed and integrated REST APIs — including ~3 third-party API integrations — to extend application functionality and streamline data exchange between systems.
- Diagnosed and resolved production bugs through systematic debugging in collaboration with a cross-functional product and engineering team of ~8, improving overall application stability by ~25%.
- Collaborated cross-functionally with frontend developers and QA teams within Agile sprints, using Git for version control to support consistent, on-schedule delivery.

Backend Developer

Apr 2023 – Feb 2025

CloudShope Technology | Noida, Uttar Pradesh

- Built and deployed a scalable Interactive Voice Response (IVR) service management platform using Python and FastAPI, handling RESTful API endpoints consumed by ~1,000+ registered users in production.
- Engineered JWT authentication and Role-Based Access Control (RBAC) across the backend system, securing API access for multiple user roles.
- Automated Customer Relationship Management (CRM) workflows using Python scripting and REST API integrations, delivering a ~40% productivity lift for the sales and operations team.
- Implemented Redis-based caching and PostgreSQL query optimisation strategies, reducing average API response time by ~35% under concurrent load.
- Containerised backend services using Docker, streamlining deployment pipelines and cutting environment setup time by ~50% across development and staging environments.
- Automated repetitive business workflows through Python scripting and collaborated cross-functionally within Agile sprints to deliver production-ready backend features on schedule.

PROJECTS

Inventory Management System — Full-Stack REST API

[Flask](#) · [React](#) · [PostgreSQL](#) · [JWT](#) · [Docker](#)

- Architected a full-stack inventory tracking application with a Flask REST API backend, JWT authentication, and a normalized PostgreSQL schema; containerized the stack with Docker for repeatable deployment.

E-Commerce Platform Backend — DRF REST API

[Django](#) · [DRF](#) · [MySQL](#) · [Redis](#) · [Razorpay](#)

- Engineered a DRF-based REST API handling product/category management, shopping cart, and order processing logic, secured end-to-end with JWT authentication.
- Integrated Razorpay for payment processing and applied Redis caching on product listing queries to reduce response times under load.

CRM Automation & Notification System

[Python](#) · [FastAPI](#) · [Redis](#) · [WhatsApp Business API](#)

- Designed and built an event-driven notification system automating multi-channel sales communication via the WhatsApp Business API, cutting manual follow-up effort.

Django REST Authentication & CRUD Platform

[Python](#) · [Django](#) · [DRF](#) · [MySQL](#) · [Swagger/OpenAPI](#)

- Built a reusable authentication and CRUD backend with JWT-based login and role-based access control (RBAC), documenting and testing REST endpoints via Swagger/OpenAPI and Postman.

DATA SCIENCE & MACHINE LEARNING PROJECTS

Employee Salary Prediction using Machine Learning

[Python](#) · [Pandas](#) · [NumPy](#) · [Matplotlib](#) · [Seaborn](#) · [Scikit-learn](#) | [GitHub](#) | [Live Demo](#)

- Built an end-to-end machine learning pipeline to predict employee salaries, performing data cleaning, missing-value imputation, duplicate removal, and outlier detection to ensure reliable model inputs.
- Conducted Exploratory Data Analysis (EDA) using Pandas, NumPy, Matplotlib, and Seaborn to analyze salary trends and key drivers across departments and experience levels.
- Engineered features using label encoding, one-hot encoding, ColumnTransformer, and feature scaling to improve model readiness and predictive accuracy.
- Trained and compared 6 regression algorithms — Linear Regression, Decision Tree, Random Forest, Gradient Boosting, Extra Trees, and XGBoost — to identify the top-performing model.
- Evaluated models using MAE, RMSE, and R² Score, selecting Extra Trees Regressor as the best-performing model with an R² score of approximately 0.68.
- Developed and deployed an interactive Streamlit web application enabling real-time employee salary predictions from the trained model.

Netflix Data Analysis

[Python](#) · [Pandas](#) · [Matplotlib](#) · [Seaborn](#) | [GitHub](#)

- Performed end-to-end exploratory data analysis on the Netflix Movies and TV Shows dataset, cleaning and standardizing raw data to uncover trends in content type, release year, ratings, genres, and country-wise distribution.
- Analyzed content distribution by country, genre, and rating, and built visualizations to communicate business insights, including the platform's shifting focus between Movies and TV Shows over time.

EDUCATION

B.Tech, Computer Science & Engineering (Pursuing)

Jul 2025 – 2028

Dr. A.P.J. Abdul Kalam Technical University (AKTU) — Modinagar Institute of Technologies

Diploma, Computer Science (BTEUP)

Government Polytechnic, Changipur

CERTIFICATIONS

- Complete 2026 Python Bootcamp: Learn Python from Scratch — Udemy (CodeWithHarry): Python, OOP, Data Structures, File Handling
- MySQL Course — Intellipaat: MySQL, SQL, Joins, Constraints, Database Design

ACHIEVEMENTS

- Performer of the Week & Helping Hand Award, CloudShope Technology — recognized for reliable sprint delivery and mentoring junior developers.